Just Ask the Model: One-Shot LLM Research Evaluation and Structured Expert Review

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Abstract

Peer review is strained, and AI tools generating referee-like feedback are already adopted by researchers and commercial services—yet field evidence on how reliably frontier LLMs can evaluate research remains scarce. We compare structured evaluations produced by six LLMs spanning different capability and cost tiers against paid expert review packages from The Unjournal, an open evaluation platform covering economics and social-science working papers, where both humans and models rate papers on seven percentile criteria with uncertainty intervals and provide narrative critiques. Treating human evaluations as a high-quality but noisy reference signal, we find that top reasoning-capable models often approach the agreement levels observed among human evaluators on several criteria, while exhibiting consistent failure modes: compressed rating scales, uneven criterion coverage, and variable identification of expert-flagged concerns. Our results suggest top reasoning models can currently serve as supplementary raters in structured evaluation pipelines, though compressed rating scales and uneven coverage of expert-flagged concerns indicate clear limitations. These are preliminary results; the paper corpus, model roster, and methods are under active development.

Introduction

Preliminary results

This paper presents first results from an ongoing study: 60 papers evaluated by six frontier LLMs against structured expert reviews from The Unjournal. The paper corpus, model roster, and methods are under active development; current and planned extensions are described in *Current work and next steps*.

A collaboration with The Unjournal (55+ open evaluation packages for global-priorities research). Funding: Survival and Flourishing Fund, Long Term Future Fund, EA Funds.

Peer review is under strain. Reviewers are hard to find, turnaround times are lengthening, and the system costs an estimated $1.5 billion per year in the United States alone (Aczel, Szaszi, and Holcombe 2021). At the same time, generative AI lowers the cost of producing polished manuscripts; in at least some fields, editors report submission growth that exceeds reviewer capacity, and explicitly link this trend to LLM-assisted writing (Spitzer 2026). This combination creates demand for automated support in editorial and pre-submission workflows.

Commercial AI reviewer products—such as Refine (Refine n.d.), IsItCredible (IsItCredible.com n.d.), and QED Science (QED Science 2026)—already market automated referee-like feedback directly to authors, though they explicitly disclaim substituting for human peer review and their internal architectures remain undocumented. OpenAIReview (Hsu and Tan 2026) takes a different approach: an open-source, transparent pipeline that uses progressive prompting to generate detailed critiques at roughly $4 per paper. Meanwhile, publishers are formalizing policies that restrict reviewer use of general-purpose AI tools while permitting controlled in-house applications for screening tasks (Elsevier 2025; Leung 2026). Our study complements these efforts by asking how far the simplest possible setup—a single prompt to a frontier model, with no bespoke pipeline—can go.

These developments make the evidentiary gap salient: funders, editors, and policymakers need to know when AI evaluation outputs are trustworthy enough to use, and when they are unstable, biased, or manipulable. Recent work documents three interlocking concerns. Reproducibility can be "jagged" across models and time (Thomas, Romasanta, and Pujol Priego 2026), and subtle task reframings can induce systematic output shifts reminiscent of specification search (Asher et al. 2026). Adversarial manipulation is not hypothetical: invisible prompt-injection text can inflate LLM review scores in simulated peer review (Choi et al. 2026). And even without manipulation, AI reviews tend to be less thematically diverse and less focused on interpretation and originality than human reviews (Rajakumar et al. 2026), while LLM scoring exhibits range restriction and halo effects that distort agreement metrics (Wang et al. 2025).

The central question we address is therefore: how reliably can frontier LLMs evaluate research, relative to expert peer review and under realistic levels of rater disagreement? We study this question in a setting designed to make "expert judgment" observable and multi-dimensional rather than implicit.

We use The Unjournal's structured human evaluations as a reference signal. We prompt six LLMs spanning different capability and cost tiers—GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Claude Opus 4.6[1], and Gemini 2.0 Flash—with the same rubric and guidelines used by human evaluators, then compare the resulting quantitative ratings and qualitative critiques against expert evaluations for 60 economics and social-science working papers. We ask whether frontier AI evaluations can approximate expert judgment, which models perform best across rating criteria and cost tiers, and whether systematic differences reveal characteristic AI preferences over research. Our headline finding is that the best-performing model (GPT-5 Pro) matches or exceeds pairwise human inter-rater rank agreement on overall quality—that is, the LLM-human correlation is comparable to the human-human correlation, which is the appropriate benchmark given substantial disagreement among human evaluators themselves. This suggests that top reasoning models can currently serve as supplementary raters in structured evaluation pipelines, even under our minimal one-shot setup.

Our approach is minimal by design: each model receives the same PDF and a fixed rubric in a single prompt, with no iteration, retrieval augmentation, chain-of-thought scaffolding, or multi-step agentic loop. This makes our results a conservative lower bound on what LLM-based evaluation can currently achieve. If frontier models already yield meaningful agreement with expert reviewers under the simplest possible setup, more sophisticated pipelines—structured measurement schemas (Asirvatham, Mokski, and Shleifer 2026), iterative quality-checking workflows (Zhang and Abernethy 2025), or the kind of prompt-robustness engineering motivated by specification-search concerns (Asher et al. 2026)—should improve further. Quantifying how much headroom remains

[1] Claude Opus 4.6 was evaluated without extended thinking enabled; results for this model reflect standard inference and likely understate its ceiling capability. A re-run with extended thinking is planned.

above this one-shot baseline, and which pipeline elements unlock it, is a key direction for future work.

The Unjournal setting is particularly well suited for this comparison. It commissions paid expert evaluations using a structured rubric covering seven percentile criteria with 90% credible intervals plus journal-tier predictions, and publishes the resulting packages openly rather than making binary accept/reject decisions—which may increase reviewer effort through accountability and transparency. The resulting ratings and critiques still exhibit substantial inter-rater variation; accordingly, we treat human evaluations as a high-quality but noisy reference signal, not ground truth. The rich, multi-dimensional data allow us to compare the priorities and calibration of humans and AI models across criteria and domains, while eventual publication outcomes for journal-tier predictions provide an external validation opportunity[2] enabling a human-vs-LLM horse race. Finally, The Unjournal's pipeline of future evaluations allows for clean out-of-training-data predictions, serving as a live testing lab for prospective validation.

Results

We evaluate 6 frontier LLMs against human expert reviews from The Unjournal. Across all models, 45 papers have both human and LLM evaluations; 33 of these were also evaluated by Claude Opus 4.6 and form the matched sample used for rank comparisons and agreement tables below. Each model receives the same PDF, system prompt mirroring The Unjournal rubric, and JSON schema requiring a diagnostic summary plus numeric midpoints and 90% credible intervals for every metric. Full methodological details appear in Methods.

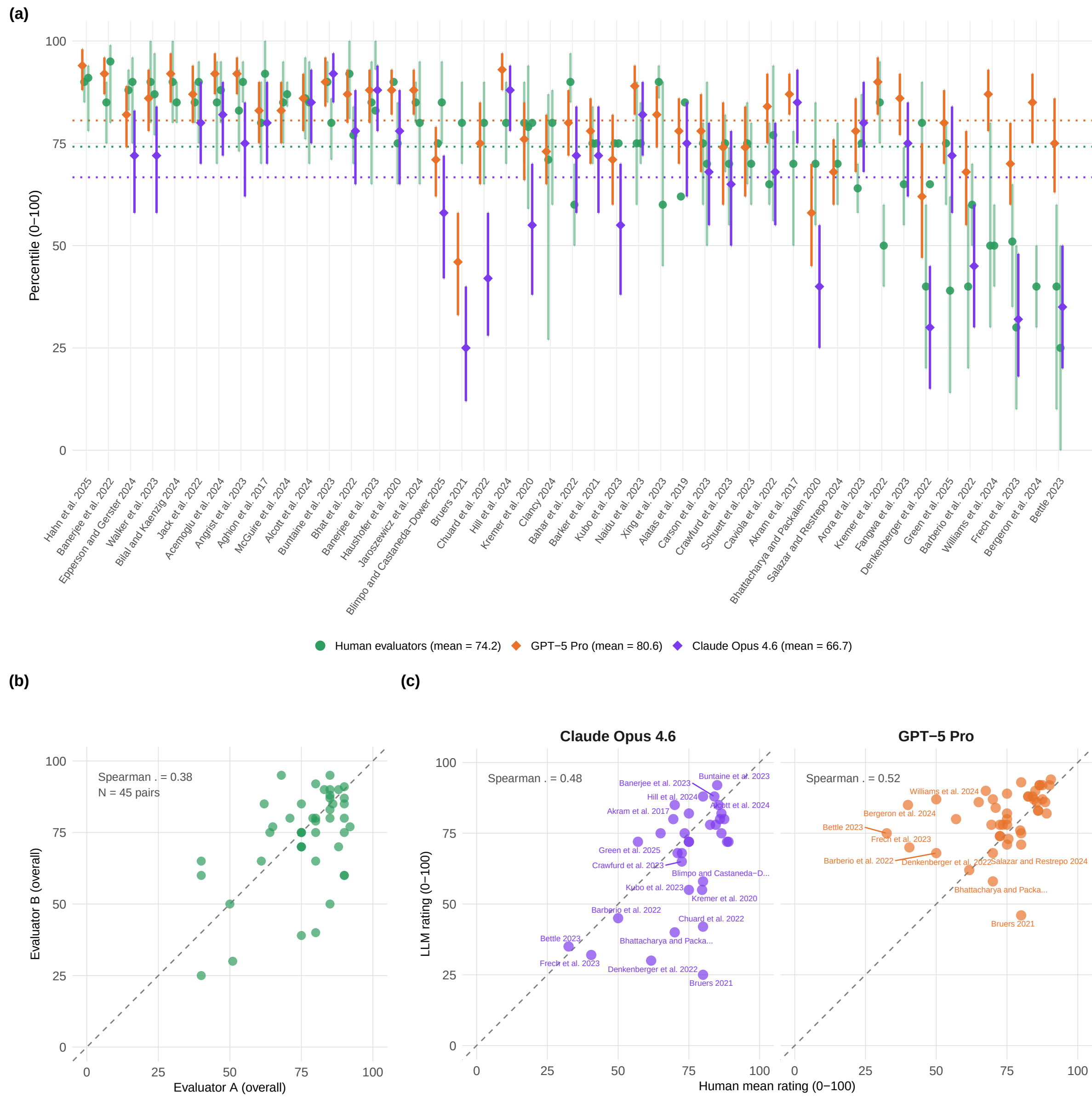
We do not treat human ratings as ground truth. Quantitative percentile scoring is genuinely difficult: even domain experts disagree, and individual scores reflect both signal about paper quality and idiosyncratic tendencies (severity, topic familiarity, interpretation of the scale). Our question is therefore whether LLMs provide signal *comparable to an additional expert rater*. If two human evaluators agree at Spearman $= 0.55$, an LLM achieving $= 0.57$ against the human mean is performing within human inter-rater range. The Human–Human baseline row in Table 1 makes this explicit; Krippendorff's $_{\mathrm{HH}}$ in the human-baseline table in Appendix A provides the criterion-level ceiling.

Per-paper overview and model comparison. Figure 1 presents three complementary views of overall (0–100 percentile) ratings. Panel (a) displays individual human evaluator ratings alongside GPT-5 Pro (orange diamonds) and Claude Opus 4.6 (purple diamonds) ratings for each paper, revealing inter-rater variability—the self-reported 90% credible intervals from individual evaluators often span 20–40 percentile points. In most cases both LLMs fall within the range of human opinions, though several papers show substantial divergence. Panel (b) compares the two highest-performing reasoning models directly against human mean ratings. Panel (c) plots evaluator pairs against each other for all papers with 2 raters, making the human-human agreement ceiling directly visible alongside LLM-human agreement in panel (b).

Panel (c) makes the human-human ceiling visual: the scatter of evaluator pairs is no tighter than the LLM-human scatter in panel (b), and individual pairs disagree by as much as 30–40 percentile points on individual papers. Note that panel (c) uses all 45 matched papers while the Human–Human row in Table 1 is restricted to the 33-paper matched sample (explaining the slightly different values). Both models cluster around the identity line in the 40–80 range but diverge more at the extremes. LLMs tend to compress ratings toward the centre of the scale relative to humans,

[2]These represent verifiable publication outcomes, not statements about the "true quality" of the paper.

Figure 1: Per-paper overall ratings (0–100 percentile). **(a)** Individual human evaluator midpoints (green circles) with each evaluator's self-reported 90% credible interval (reflecting their own uncertainty about the true score; the vertical separation between green dots per paper reflects inter-rater disagreement), GPT-5 Pro (orange diamond with CI), and Claude Opus 4.6 (purple diamond with CI), sorted by descending human mean. Dotted horizontal lines show grand means. **(b)** Pairwise human evaluator agreement across all matched papers: each point is one evaluator pair (papers with 3 raters contribute 3 points); uses all `r n_matched` matched papers (the Human–Human row in Table 1 is restricted to the `r n_opus`-paper matched sample). **(c)** Human mean vs LLM overall rating for the two focal models with per-model Spearman annotated; dashed diagonal is the identity line. Compare panels (b) and (c) directly to see whether LLM-human scatter is tighter than human-human scatter.



producing narrower spread—a pattern consistent with alignment training that discourages extreme outputs. Where humans rate a paper very highly or very harshly, the LLM typically pulls toward the middle. Full agreement metrics including Pearson r, Spearman , mean bias, and RMSE for all six models appear in the agreement table in Appendix A.

Rank comparison. Figure 2 places human rankings in the centre column with GPT-5 Pro on the left and Claude Opus 4.6 on the right, restricted to the 33 papers with Claude Opus 4.6 evaluations. Each paper appears as a pair of S-curves connecting its human rank to its rank under each LLM. Horizontal curves indicate agreement; curves that climb toward the top indicate the LLM ranked the paper higher (better) than humans did. Colour encodes the rank shift: green when humans rank it higher, orange (left, GPT-5 Pro) or purple (right, Opus) when the LLM does. The triptych layout makes it easy to spot papers where both LLMs agree with each other but disagree with humans (both curves slope the same way), versus papers where the two LLMs diverge.

Papers where both curves slope the same direction indicate systematic human–LLM disagreement; papers where left and right curves diverge indicate model-specific differences—GPT and Claude "see" the paper differently, not just differently from humans. Horizontal curves indicate agreement with the human ordering. The criteria-level Krippendorff's alpha analysis in the human baseline table in Appendix A quantifies where human–LLM agreement approaches the ceiling set by human inter-rater variability.

Rating differences by paper and criterion. While the scatter plots and slopegraph summarise agreement on a single dimension, Figure 3 unpacks Human − LLM differences across all seven criteria simultaneously. Each column is a paper, each row a criterion, and tile colour encodes the signed difference (human mean minus LLM midpoint). Green tiles indicate the human panel rated the paper higher; orange (GPT-5 Pro) or purple (Claude Opus) tiles indicate the LLM was more generous. Papers are sorted by GPT-5 Pro's overall difference so that both panels share the same column order, making model-specific patterns immediately visible.

Rows with uniformly green or orange tiles indicate papers where humans and the LLM disagree systematically across all criteria, not just on one dimension. Columns with consistent colour suggest criteria-level biases—for instance, if "Open Science" is green across nearly all papers, humans may systematically reward data-sharing practices more than the LLM does. Comparing the two panels reveals whether disagreement patterns are model-specific or shared across frontier LLMs; tiles that change colour between panels highlight papers where the models diverge from each other, not just from humans.

Qualitative critique comparison. Beyond numeric ratings, we compare the substantive critiques each model raises against the consensus issues identified by human experts. The figure below plots coverage—the fraction of human-identified concerns that the LLM also raised in some form—against precision—the fraction of LLM-raised issues that correspond to a substantive human concern. These metrics are assessed by GPT-5.2 Pro acting as an independent judge (see Methods for the judging protocol).

Coverage and precision vary substantially across papers. Some papers achieve high scores on both dimensions, indicating strong alignment between human and LLM critiques, while others reveal the LLM missing key human concerns or raising issues absent from the expert consensus. Because these metrics are themselves LLM-assessed, they should be interpreted with the caveat that an LLM judge may systematically over- or under-credit matches relative to a human annotator; manual validation through our annotation tool is underway and preliminary results are consistent with the automated scores.

Figure 2: Triptych slopegraph of overall ratings for the `r n_papers_slope`-paper matched sample (papers evaluated by all three sources). Each column shows rank order (1 = highest rated) within that source; papers are connected by S-shaped curves. Green = human panel ranks higher; orange (left) = GPT-5 Pro ranks higher; purple (right) = Claude Opus ranks higher. Colour intensity encodes magnitude of rank shift. Right-side labels show human rank (H) and rank delta for each model (ΔG, ΔC; positive = LLM ranks higher).

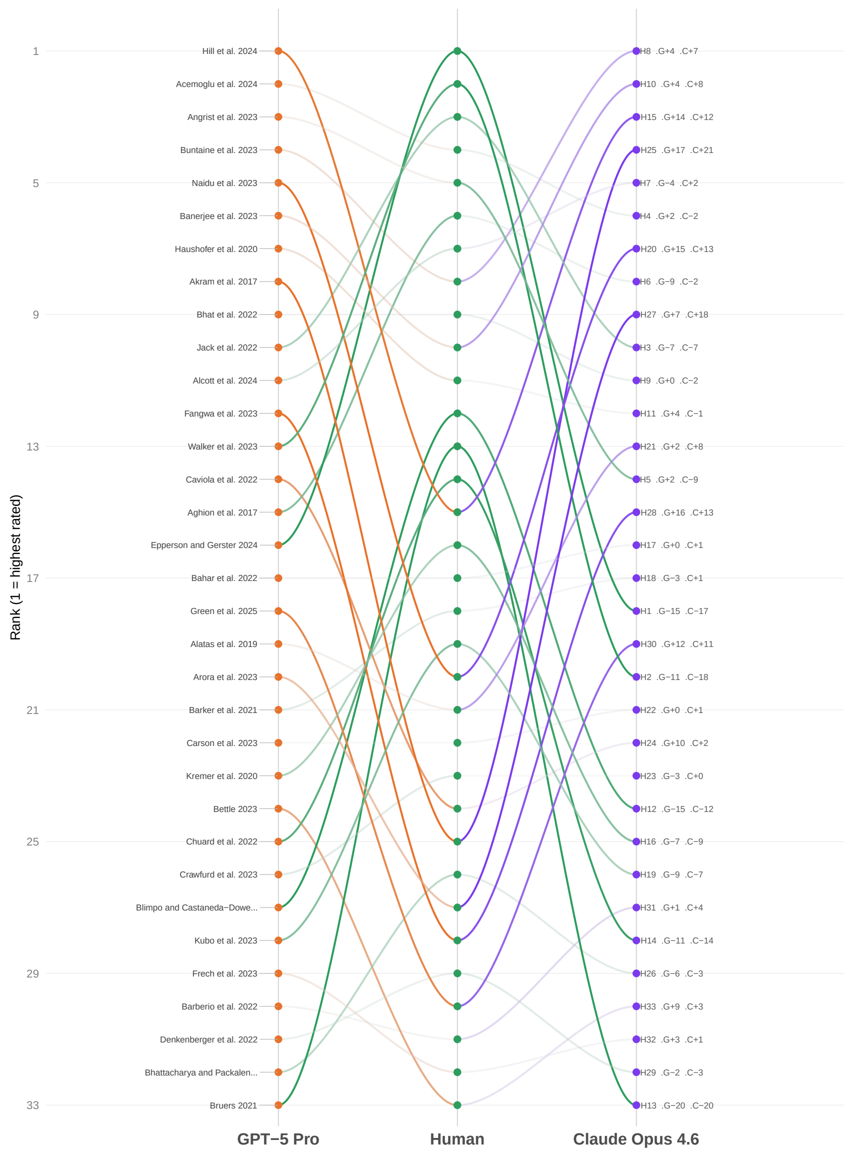
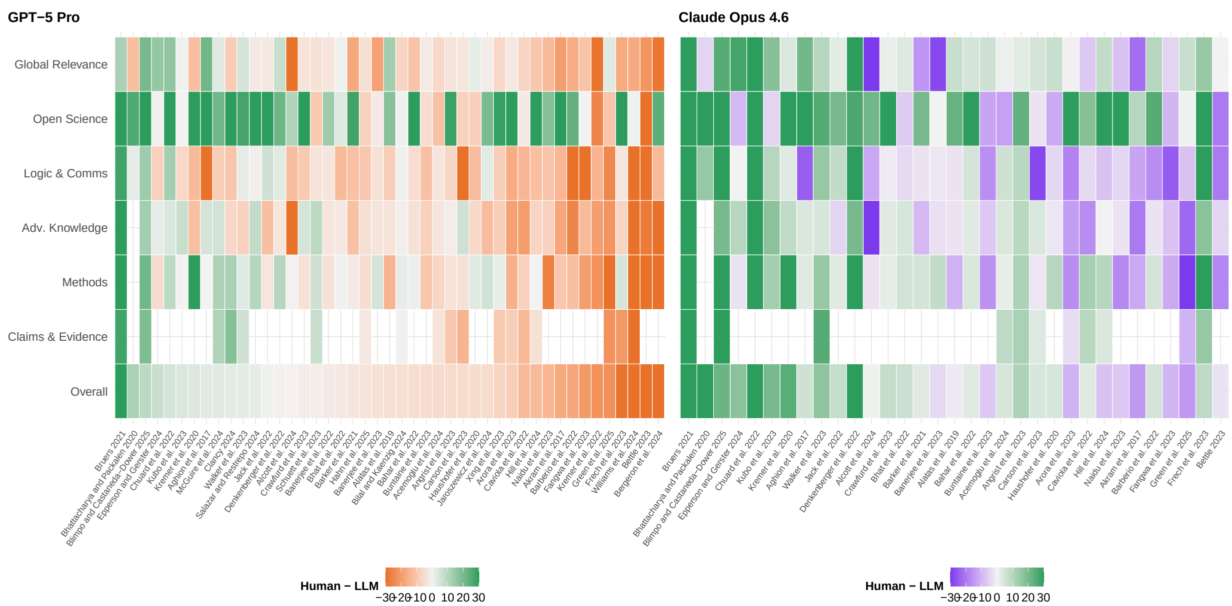


Figure 3: Human minus LLM rating difference for every paper (columns) and criterion (rows), comparing GPT-5 Pro (left panel) and Claude Opus 4.6 (right panel). Green tiles indicate the human mean was higher; orange (GPT-5 Pro) or purple (Claude Opus) tiles indicate the LLM rated the paper higher. Colour is clamped to ±30 percentile points. Papers are sorted by GPT-5 Pro overall difference so the two panels are directly comparable.



Detailed paper-by-paper comparisons—including matched issue pairs with severity labels, structural difference tables, and per-evaluator breakdowns—appear in Appendix B: Critiques & Key Issues. Extended quantitative analysis, including per-criterion correlations, bootstrap confidence intervals, mutual information, tier prediction accuracy, cost-quality trade-offs across all six models, and top disagreement cases, is reported in Appendix A: Results Ratings. The full LLM reasoning traces and assessment summaries are available in Appendix C: LLM Traces.

Model comparison summary. Table 1 compares all six models on overall-rating agreement metrics against the human mean, restricted to the 33-paper matched sample so every model is evaluated on identical papers. The **Human–Human** row is the empirical ceiling: it shows pairwise agreement between evaluators on the same papers. Models are sorted by Spearman .

Table 1: Agreement between each LLM and human mean overall rating (matched sample, N = `r n_opus` papers). **Human–Human** row (top, bold) shows pairwise evaluator agreement as an empirical upper bound. Spearman (rank-based) and MAE (in percentile points, directly interpretable) are primary metrics; Pearson r shown for completeness. Bias = LLM − Human (positive = LLM more generous). Models sorted by Spearman .

Model	N	Spearman	Pearson r	Mean bias	RMSE	MAE
Human–Human	**27**	**0.472**	**0.584**	**+3.5**	**15.6**	**11.3**
GPT-5 Pro	33	0.553	0.396	+5.1	14.1	10.0
Claude Opus 4.6	33	0.477	0.567	-7.6	17.2	12.4
Gemini 2.0 Flash	33	0.447	0.467	-5.3	13.2	10.8
GPT-5.2 Pro	5	0.410	0.433	-6.6	19.9	15.4

GPT-4o-mini	33	0.309	0.356	-0.1	15.9	10.7
Claude Sonnet 4	8	-0.058	-0.232	-1.6	9.1	6.8

Table 2 breaks the same agreement metrics down by evaluation criterion, averaged across all models. Crucially, the **H-H** column shows pairwise human-human Spearman for each criterion — the ceiling available to any model. Criteria where H-H is itself low indicate genuine expert disagreement; low LLM agreement on those criteria is therefore expected rather than a model failure.

Table 2: Human–LLM agreement by criterion, averaged across all models (matched sample, N = `r n_opus` papers). **H-H** shows pairwise human evaluator Spearman (the ceiling for each criterion); LLM Spearman and other metrics are averages across all six models. Positive bias = LLMs rate higher on average. Note the ceiling variation across criteria: human evaluators agree strongly on some dimensions and barely at all on others (Open Science), which sets expectations for LLM performance.

Criterion	H-H	LLM Spearman	LLM Pearson r	Mean bias	RMSE
Overall	**0.472**	0.406	0.415	-2.1	15.1
Claims & Evidence	**0.087**	0.279	0.362	-4.5	16.9
Methods	**0.178**	0.323	0.304	-1.2	18.4
Adv. Knowledge	**0.026**	0.218	0.230	+1.1	16.8
Logic & Comms	**0.291**	0.041	0.082	+3.9	16.9
Open Science	**0.238**	0.008	0.037	-11.7	27.0
Global Relevance	**0.201**	0.257	0.181	-0.3	17.6

All appendices, interactive figures, and full code are available at the companion website: https://llm-uj-research-eval.netlify.app. Source code and data are hosted at https://github.com/valentinklotzbuecher/llm-uj-research-eval.

Discussion

Frontier LLMs achieve moderate-to-strong agreement with human expert evaluators, approaching the agreement levels observed among human evaluators themselves. Correlations with human mean ratings are highest for reasoning-capable models (GPT-5 Pro, Claude Opus 4.6), though even lightweight models (GPT-4o-mini, Gemini 2.0 Flash) achieve moderate correlations (r 0.5–0.6) at a fraction of the cost. Central compression—the tendency for LLMs to pull extreme ratings toward the middle of the scale—is the most consistent pattern across all models, likely reflecting alignment training that discourages confident extreme outputs. Qualitative coverage varies widely across papers: on some, the LLM captures nearly all consensus human concerns; on others, it misses key critiques or raises issues absent from the expert consensus.

Limitations. Several caveats temper these conclusions. Our sample comprises roughly 50 social-science papers specifically selected by The Unjournal for evaluation, not a random draw from the research literature; performance may differ in other fields or on less polished manuscripts. Human evaluations are themselves a noisy reference signal rather than ground truth, with substantial inter-rater variation that caps achievable agreement. We cannot fully rule out knowledge contamination: while we instruct the model to ignore prior knowledge about authors, institutions, or publication history in the system prompt, the models' training data may include fragments of these papers or related discussions. Robustness checks with models whose training cutoffs predate the papers, or out-of-time validation on papers entering The Unjournal's pipeline after model training, would help address this concern. Alignment training likely contributes to score inflation and narrower credible intervals than humans provide. All LLM evaluations are single-run; aggregating across multiple runs or temperature settings could change the picture. Finally, the qualitative coverage and precision metrics are themselves LLM-assessed (GPT-5.2 Pro as judge), introducing a further layer of model dependence.

Implications. The cost structure is striking: lightweight models deliver evaluations at under a cent per paper, while reasoning-capable models cost several dollars—still orders of magnitude cheaper than human expert review. This suggests a practical tier structure in which cheap models could serve as rapid screening tools and expensive reasoners could provide deeper assessment where warranted. However, the qualitative gaps we observe—missed critiques, generic issues, and central compression of ratings—argue against full automation of peer review. AI evaluation appears most promising as a supplement: providing fast structured feedback, flagging potential concerns for human reviewers, and enabling systematic comparison across large paper sets that would be infeasible with human effort alone.

Governance and attack surface. As AI review tools move from research prototypes to deployed products, the attack surface expands. Prompt-injection techniques—embedding hidden instructions in a manuscript's metadata, footnotes, or even white-on-white text—could steer model outputs toward inflated ratings or suppressed critiques. Because our pipeline (and similar commercial services) routes unpublished manuscripts through third-party APIs, confidentiality cannot be guaranteed without end-to-end encryption or on-premise deployment. Over-reliance on AI scores introduces a further governance risk: if editorial decisions weight model ratings, authors may optimise papers for the model rather than for scientific rigour, creating a Goodhart dynamic. Finally, current evaluations reflect a single model checkpoint; model updates, alignment changes, or fine-tuning can shift ratings in ways that are invisible to users. We recommend that any operational deployment include adversarial red-teaming of prompts, formal confidentiality agreements with API providers, transparent disclosure of AI involvement in review, and periodic re-calibration against fresh human evaluations.

Current work and next steps. {#current-work-and-next-steps} This paper reports first results; several extensions are in progress.

Out-of-time validation. We are expanding the paper sample with Unjournal evaluations entering the pipeline after model training cutoffs, eliminating contamination concerns and enabling prospective performance tracking across domains.

Robustness and prompt sensitivity. Systematic prompt-variation and multi-run experiments are underway to quantify how sensitive LLM ratings are to prompt framing—a practical concern highlighted by recent work on LLM specification search (Asher et al. 2026).

Human validation of qualitative metrics. The coverage and precision metrics reported here are themselves LLM-assessed. We are designing human-enumerator studies to independently validate whether LLMs identify the same expert-flagged concerns.

Journal-outcome horse-race. A calibration benchmark comparing human and LLM tier predictions against verified publication venues is planned, offering a rare opportunity for externally verifiable accuracy measurement.

Further extensions include content-swap bias tests (Pataranutaporn et al. 2025) to probe author- and institution-driven biases, agentic or multi-step evaluation pipelines, and hybrid human-AI evaluation trials.

Methods

This chapter describes the data sources, evaluation pipeline, prompt design, and statistical methods used throughout. All Python code underlying the pipeline is preserved in the collapsible blocks below and can be executed to reproduce the evaluation runs.

Sample and human reference data. We draw on published evaluations from The Unjournal, an open-access platform that commissions expert reviews of policy-relevant research without requiring journal submission. Each paper is typically assessed by two independent evaluators (occasionally one or three), who provide a written critique resembling a referee report together with quantitative ratings on seven criteria scored as percentiles (0–100) relative to "all serious research in the same area encountered in the last three years."[3] Evaluators additionally predict the journal tier in which the work "should" and "will" publish, using a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable, 5 = top-5 journal). For each metric, evaluators report a midpoint (median of their belief distribution) and a 90% credible interval that expresses their epistemic uncertainty.

The sample comprises working papers spanning 2017–2025 in development economics, health policy, environmental economics, and related fields that The Unjournal identified as high-impact. All papers have completed The Unjournal's full evaluation process, meaning the authors received evaluations that have been publicly posted. For our analysis we extracted the individual evaluator scores, aggregated them by taking the arithmetic mean per paper per criterion, and retained the range of individual scores to characterise inter-rater spread.

LLM evaluation pipeline. We evaluate six frontier models spanning three providers: GPT-5 Pro and GPT-5.2 Pro (OpenAI, reasoning-capable), GPT-4o-mini (OpenAI, lightweight), Claude Sonnet 4 and Claude Opus 4.6 (Anthropic), and Gemini 2.0 Flash (Google). Each model receives the identical PDF file, system prompt, and output schema. We pass the PDF directly to the model's native multimodal input rather than extracting text, preserving tables, figures, equations, and layout cues that ad-hoc scraping could mangle. A single API call per paper avoids hand-offs and summary loss from multi-stage pipelines.

The output is constrained by a strict JSON Schema enforcing the same nine fields that human evaluators complete: seven percentile metrics (each with `midpoint`, `lower_bound`, `upper_bound`) and two journal-tier predictions (each with `score`, `ci_lower`, `ci_upper`). Additionally, the model produces an `assessment_summary` of approximately 1,000 words that must precede scoring—a "think first, score second" protocol designed to ground numeric ratings in specific textual evidence. The extended schema used for our GPT-5.2 Pro focal run adds a `key_issues` array of concise, ranked issue statements for downstream critique comparison.

Agreement and reliability statistics. We quantify human–LLM agreement using several complementary measures. *Pearson's r* captures linear association between paired ratings and is sensitive to proportional biases but not to constant offsets. *Spearman's* measures rank-order agreement and is robust to outliers and non-linear monotone relationships. *Mean bias* (LLM minus human) indicates the direction and magnitude of any systematic offset. *Root mean squared error* (RMSE) and *mean absolute error* (MAE) measure typical prediction error in the original scale units; RMSE penalises large deviations more heavily. To contextualise human–LLM agreement we report *Krippendorff's alpha* (α), a chance-corrected reliability coefficient that generalises across varying numbers of raters, accommodates missing data, and applies to any measurement level (nominal, ordinal, interval, or ratio). An α of 1 indicates perfect agreement, 0 indicates agreement no better than chance, and values below 0 indicate systematic disagreement. We compute α_{HH} (among human evaluators only) as a ceiling: if human evaluators agree with each other at only $\alpha = 0.5$ on a given criterion, expecting an LLM to exceed that level would be unrealistic. We then report α_{HL} (between the human mean and each LLM) to assess how close machine ratings come to this ceiling. For the qualitative key-issue comparison, *coverage* denotes the fraction of human-identified issues that received a match score $\geq$ 30% from the LLM judge, and *precision* denotes the fraction of LLM-generated issues that matched at least one human issue.

[3]The seven criteria are: *overall assessment*, *claims and evidence*, *methods*, *advancing knowledge and practice*, *logic and communication*, *open and collaborative science*, and *relevance to global priorities*. Full definitions are given in The Unjournal's guidelines for evaluators.

JSON Schema. The output schema enforces that every paper is scored on identical fields with identical types and bounds. Credible intervals are required (paralleling the human protocol) so that the model can express genuine uncertainty rather than suggest false precision.

System prompt design. The system prompt is assembled from modular components and concatenated before each API call. It opens with a role definition instructing the model to act as an expert research evaluator, followed by a debiasing block that explicitly prohibits use of author identity, institutional prestige, publication venue, or any extrinsic information—the model must base all judgments on the PDF content alone. A diagnostic-summary instruction requires the model to produce a roughly 1,000-word assessment identifying methodological, evidential, and interpretive issues *before* any scoring, implementing a "think first, score second" protocol intended to anchor numeric ratings in specific textual evidence.

Percentile ratings are anchored to the reference group "all serious research in the same area encountered in the last three years," following The Unjournal's guidelines for evaluators. The prompt defines each of the seven criteria with emphasis on global priorities and practical relevance over pure academic novelty, mirroring the weight structure that human Unjournal evaluators are asked to apply.

Subsequent prompt components instruct the model on constructing 90% credible intervals—the smallest interval the evaluator believes is 90% likely to contain the true value—encouraging calibrated uncertainty rather than artificially narrow bounds. The prompt requests journal-tier predictions on a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable through 5 = top-5 journal), providing an externally verifiable reference point for papers that eventually publish. A validation block then requires the model to verify internal consistency: numeric scores must align with the written assessment, credible intervals must be non-degenerate, and high or low ratings must be explicitly justified in the assessment summary.

The components above are concatenated into a single system prompt string before each API call:

Submission and collection. The `evaluate_paper` function uploads a PDF to the API and submits a background job for evaluation. File IDs are cached by path, size, and modification time so that re-running on the same PDF reuses the previously uploaded file.

Each model receives the full PDF in a single reasoning call, avoiding hand-offs and summary loss from multi-stage ingestion. We submit one background job per paper to the OpenAI Responses API with "high" reasoning effort and server-side JSON-Schema enforcement, recording the response ID, model, file ID, status, and timestamps in a jobs index. No external sources or cross-paper material are retrieved; the evaluation is anchored entirely in the manuscript itself.

A polling loop then checks each job's status and, for completed jobs, retrieves the raw JSON response object and writes it to disk alongside reasoning-trace metadata (token counts, reasoning summary).

Multi-model evaluation. To assess whether systematic biases or calibration differences emerge across architectures, we collect evaluations from six models spanning three providers. For OpenAI models that lack background-job or extended-reasoning support (e.g., GPT-4o-mini), we use a synchronous call variant:

Anthropic. Anthropic's API accepts PDFs as base64-encoded document content rather than uploaded file IDs, and all calls are synchronous. We use Claude's native PDF support to preserve the same multimodal evaluation approach:

Google. Google's Gemini API accepts PDFs via a file-upload endpoint with MIME-type tagging:

A unified runner dispatches evaluations across all configured providers and writes per-model output directories:

Focal run with key-issue extraction. For a subset of 14 papers with rich human critiques, we ran an extended evaluation using GPT-5.2 Pro. The schema for this focal run adds a `key_issues` array—a ranked list of concise issue statements identifying the most important methodological, interpretive, or evidential concerns—alongside the standard metrics. This structured output enables direct comparison between machine-generated and human-identified issues.

Key-issue comparison with human critiques. To validate how well the LLM identifies substantive concerns, we compare its `key_issues` output against human expert critiques drawn from The Unjournal's Coda database. These human critiques—produced by paid domain experts and synthesized by evaluation managers—provide a high-quality but noisy reference standard for issue identification; accordingly, we treat them as a comparative signal rather than ground truth.

The comparison proceeds in two stages. First, we parse and align the data sources: LLM key issues are extracted from the focal-run JSON responses, while human critiques are drawn from a manually curated markdown document pairing each paper's machine and expert assessments. Second, we use an LLM judge (GPT-5.2 Pro with schema-enforced structured output) to systematically assess the degree of alignment between each human-identified issue and the set of machine-generated issues, producing issue-by-issue match scores, coverage estimates (fraction of human issues captured), and precision estimates (fraction of LLM issues that are genuinely substantive).

The comparison pipeline takes a manually curated markdown file pairing each paper's LLM issues with human critiques and produces two structured outputs: a parsed JSON dataset (`key_issues_comparison.json`) and an LLM-assessed alignment report (`key_issues_comparison_results.json`) containing per-paper coverage, precision, matched-pair explanations, and an overall quality rating. Together, these outputs feed the qualitative analysis presented in Appendix B: Critiques & Key Issues.

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Results: Ratings

Extended ratings analysis (per-criterion correlations, agreement metrics, human baseline, cost–quality trade-offs) is available in the online supplement at https://llm-uj-research-eval.netlify.app/results_ratings.html.

Results: Critiques & Key Issues

Detailed paper-by-paper critique comparisons (matched issue pairs, severity labels, structural differences) are available in the online supplement at https://llm-uj-research-eval.netlify.app/results_critiques.html.

Supplementary Material

Full LLM reasoning traces, assessment summaries, related work, and extended analyses are available in the online supplement at https://llm-uj-research-eval.netlify.app/appendix_llm_traces.html.